

ITA-1407 ETHICAL HACKING

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1. A mid-sized financial organization operates a Windows domain network consisting of 150 interconnected systems. During a security incident investigation, it is observed that attackers have exploited NULL sessions, open RPC services, and misconfigured shared folders to gain unauthorized access. Initial analysis indicates that 45% of systems allow anonymous enumeration, and among those, 60% are susceptible to SMB-based lateral movement attacks. Furthermore, each compromised system can attempt to access 3 additional systems, with a success probability of 0.3 per attempt.

Answer

Given:

- Total systems = 150
- Anonymous enumeration = 45%
- SMB vulnerable systems = 60%
- Additional attempts per compromised system = 3
- Success probability = 0.3

Calculations:

Systems allowing anonymous enumeration:

$$150 \times 0.45 = 67.5$$

Initially compromised systems:

$$67.5 \times 0.60 = 40.5$$

Lateral movement compromises:

$$40.5 \times 3 \times 0.3 = 36.45$$

Total compromised systems:

$$40.5 + 36.45 = 76.95 \approx 77$$

Percentage compromised:

$$(77 \div 150) \times 100 = 51.3\%$$

Result: Approximately **77 systems** may be compromised, representing about **51.3% of the network**. Since this is below **70%**, the network is not at immediate risk of a large-scale compromise. **NULL sessions** should be prioritized for mitigation because they enable the initial unauthorized access.

2. An enterprise network consists of 120 systems, out of which 70 systems expose RPC services to the internal network. A vulnerability scan reveals that 40% of these systems are susceptible to known exploits, and the probability of successful exploitation for each vulnerable system is 0.5. However, each exploitation attempt carries a detection probability of 0.25 due to intrusion detection systems deployed within the network.

Answer

Given:

- RPC-enabled systems = 70
- Vulnerable systems = 40%
- Exploitation success probability = 0.5
- Detection probability = 0.25

Calculations:

Vulnerable systems:

$$70 \times 0.40 = 28$$

Successfully compromised systems:

$$28 \times 0.5 = 14$$

Detection alerts:

$$14 \times 0.25 = 3.5$$

Result: The attacker can successfully compromise approximately **14 systems**, while generating around **3.5 detection alerts**. Due to the risk of detection, a stealthy attacker may prefer alternative attack methods instead of extensive RPC exploitation.

3. A cloud-hosted Linux server is running multiple services exposed to the internet. Security analysis identifies four critical services, with independent probabilities of vulnerability being 0.3, 0.2, 0.25, and 0.15, respectively. It is also observed that the system has weak file permissions, which can be exploited with a probability of 0.5 once initial access is gained.

Answer

Given:

- Service vulnerability probabilities = 0.3, 0.2, 0.25, 0.15
- Privilege escalation probability = 0.5

Calculations:

Probability that no service is vulnerable:

$$\begin{aligned}(1 - 0.3) \times (1 - 0.2) \times (1 - 0.25) \times (1 - 0.15) \\= 0.7 \times 0.8 \times 0.75 \times 0.85 \\= 0.357\end{aligned}$$

Probability that at least one service is vulnerable:

$$\begin{aligned}1 - 0.357 = 0.643 \\= 64.3\%\end{aligned}$$

Probability of full compromise:

$$\begin{aligned}0.643 \times 0.5 = 0.3215 \\= 32.15\%\end{aligned}$$

Result: The probability that at least one service is vulnerable is **64.3%**, while the probability of a full system compromise is **32.15%**. Strengthening file permissions can significantly reduce the overall risk of compromise.

4. An organization uses Network File System (NFS) to share resources across three internal subnets. Due to improper configuration, unauthorized access probabilities from these subnets are estimated as 0.4, 0.3, and 0.2, respectively. Once access to the NFS share is obtained, the probability of privilege escalation to root access is 0.7.

Answer

Given:

- Subnet A = 0.4
- Subnet B = 0.3
- Subnet C = 0.2
- Privilege escalation probability = 0.7

Calculations:

Probability that no subnet gains access:

$$\begin{aligned}(1 - 0.4) \times (1 - 0.3) \times (1 - 0.2) \\ = 0.6 \times 0.7 \times 0.8 \\ = 0.336\end{aligned}$$

Probability that at least one subnet gains access:

$$\begin{aligned}1 - 0.336 = 0.664 \\ = 66.4\%\end{aligned}$$

Probability of complete compromise:

$$\begin{aligned}0.664 \times 0.7 = 0.4648 \\ = 46.48\%\end{aligned}$$

Result: The probability of unauthorized NFS access is **66.4%**, and the probability of complete system compromise is **46.48%**. Restricting NFS access to authorized subnets would effectively reduce the attack surface and improve security.

5. An e-commerce platform maintains a database containing 50,000 user records, of which 30% are classified as sensitive. A successful SQL injection attack allows data extraction at a rate of 500 records per second. However, the application firewall detects and blocks the attack after 40 seconds.

Answer

Given:

- Total records = 50,000
- Sensitive records = 30%
- Extraction rate = 500 records/second
- Detection time = 40 seconds

Calculations:

Total records extracted:

$$500 \times 40 = 20,000$$

Percentage of database compromised:

$$(20,000 \div 50,000) \times 100 = 40\%$$

Sensitive records exposed:

$$20,000 \times 0.30 = 6,000$$

Result: The attacker can extract approximately **20,000 records (40% of the database)** before being blocked. Around **6,000 sensitive records** may be exposed, indicating a **high-severity breach**. Additional preventive security measures are required to reduce data exposure.